Video Walkthrough Script

Using AI to Debug Without Cheating

Walkthrough Script

AI should help students code better, not avoid learning code.
AI should help students think better, not think less.

Standards summary: This resource may support CSF 9.2, CSF 13.1, CSF 16.1, CSF 16.2 when used as part of AI literacy, computer science, programming, digital ethics, or cybersecurity instruction. Detailed standards connection appears at the end of this document.

Opening

Learning to code can be frustrating because one small mistake can stop a whole program from working. AI can help students work through those moments, but only if we teach them to use it as a coach rather than a replacement coder.

Core Message

I teach students that AI can explain code, interpret error messages, offer hints, and help them practice. But students still need to read the code, make decisions, test the result, and explain what changed.

Show the Starter Kit

Lesson Plan: AI as a Coding Coach

Python Debugging Activity

Ask AI Better Coding Questions handout

Responsible AI Use in Coding Rubric

Student-safe AI coding conversation sample

Classroom Example

Instead of asking AI to fix a Python assignment, a student can ask: Explain what this error means and point me to the line I should check first. That prompt keeps the student in the learning process. The student still makes the fix, tests the code, and explains the result.

Administrator Connection

For schools, this approach gives teachers a practical way to address academic integrity without pretending students will never encounter AI. It supports responsible technology use while building real programming habits: debugging, testing, iteration, and documentation.

Closing

The goal is not to keep students away from powerful tools. The goal is to teach them how to use those tools with judgment. Students should not need AI to code for them. They should learn how to code better with it.

Optional On-Screen Text

AI as coach, not replacement

Ask for hints before fixes

Test and verify your code

Explain what changed

Never submit code you cannot explain

Detailed Tennessee Standards Connection

Use this page as implementation guidance. Alignment depends on local district pacing, approved course placement, teacher directions, and how the resource is used in instruction.

Tennessee Standards Connection

This video script supports standards by explaining how AI can support programming, debugging, ethical use, verification, and student ownership.

Standards source: Tennessee Department of Education, Computer Science Foundations (C10H11), May 2023. Confirm final alignment against local district pacing, approved course placement, and teacher directions.

This resource may support the following Tennessee standards when used as part of AI literacy, computer science, programming, digital ethics, or cybersecurity instruction:

- CSF 9.2 - Troubleshooting Process: Students use a structured process to identify a problem, gather information, isolate causes, test a solution, verify the result, and document what they learned.
- CSF 13.1 - Social, Legal, and Ethical Issues: Students identify responsibilities related to ethical technology use, academic integrity, copyright, appropriate AI use, and responsible programming support.
- CSF 16.1 - Programming Language: Students explore programming languages such as Python and explain how programmers use them to solve a variety of IT problems.
- CSF 16.2 - Software Development Life Cycle: Students connect planning, coding, testing, refinement, deployment, and maintenance to an iterative software-development process.